LESSON 1.1

STATISTICS AND INTERPRETING DATA

LESSON INTRODUCTION

6.DP.1.1 Recognize and formulate a statistical question that would generate numerical data.

6.DP.1.4 Given a histogram or line plot within a real-world context, qualitatively describe and interpret the spread and distribution of the data, including any symmetry, skewness, gaps, clusters, outliers, and the range.

LEARNING TARGETS

- I can identify statistical questions and classify types of data.
- I can interpret information from a data set or graphical display.

BENCHMARK COHERENCE

BUILDING ON

MA.5.DP.1.2 Interpret numerical data, with whole-number values, represented with tables or line plots by determining the mean, mode, median, or range.

WORKING TOWARD

MA.7.DP.1.1 Determine an appropriate measure of center or measure of variation to summarize numerical data, represented numerically or graphically, taking into consideration the context and any outliers.

ESSENTIAL QUESTIONS

- What is a statistical question?
- What are the characteristics of a data set that are displayed on a histogram or a line plot?
- How can I use the characteristics of a data set displayed on a graph to interpret the information?

LESSON NARRATIVE

Lesson 1 begins with students recalling the definition of a statistical question and determine whether questions are statistical or not. Students then transition to activating prior knowledge on histograms, box plots, and line plots.

COMPONENT SUMMARY

1.1.1 WARM-UP *(~5 MIN.)*

Day in the Life

1.1.2 REFRESHER *(5 MIN.)*

Statistical questions in review

1.1.3 TRY IT *(5 MIN.)*

Determine whether a question is a statistical question or not

1.1.4 REFRESHER *(5 MIN.)*

Review line plot, box plot, and histogram

1.1.5 GUIDED INSTRUCTION *(10 MIN.)*

Interpret histograms and line plots

1.1.6 COLLABORATION *(10 MIN.)*

Interpret and analyze a given data set displayed using two different graphs

1.1.7 WRAP-UP *(~5 MIN.)*

Self-assessment of today's learning

RESOURCES

GLOSSARY

Key Terms:

- Statistical Question

Additional Terms:

- Histogram, line plot, box plot
- Skewed, outlier, range

MATERIALS

- (Optional) Career Profile video
- (Optional) Small white board and marker for each student

ADDITIONAL PREPARATION

- Warm-Up 1.1.1 contains a video to accompany the question prompts.

TEACHER TIPS

COMMON MISCONCEPTIONS

- Students might determine that the histogram is left-skewed because most of the data falls to the left of the center.
- Students may present the range of data as an actual range, like “between 80 and 115” rather than a single number that represents the difference between the two values.

THINGS TO CONSIDER

- If this is your first lesson of the year, utilize this lesson to establish and facilitate an equitable learning environment in which all students have access to the content, to you as the teacher, and to each other. Lessons on statistics more often place students at the same “level,” rather than benchmarks that heavily rely on students who have learned and who have not learned, often starting off the school year by separating students who “have” and students who “have not.” Build equity in who is engaging, how they engage, and success for all.

LITERACY AND LANGUAGE ROUTINES

Compare and Connect

Question 1a in 1.1.6 Collaboration asks students to identify which graph better shows “how many.” Consider using the Compare and Connect routine to engage students in thinking and talking about the graphs by modeling a Think Aloud, asking questions that highlight characteristics of each graph that can help students to interpret and analyze data.

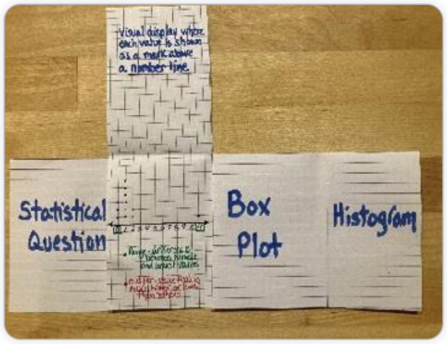
MATHEMATICAL THINKING AND REASONING (MTR) STANDARD

MA.K12.MTR.2.1 Multiple Representations

Clarifications in MTR 2 ask students to make connections between concepts and representations. Prompt students to support their thinking throughout 1.1.5 Guided Instruction and 1.1.6 Collaboration by continually directing them to state how the graph supports the answers they’ve chosen when working through the questions.

DIFFERENTIATE INSTRUCTION

This lesson offers two opportunities for review, where students are revisiting concepts already learned, to offer entry points for learners by activating prior knowledge that is necessary for the content presented in the lesson. The lesson is rich in vocabulary, and so considering the use of a foldable or some other visual for students to keep track of the vocabulary might be helpful.



1.1.1 WARM-UP: DAY IN THE LIFE OF A MUSHROOM FARMER

TEACHER MOVES

Play the Career Profile video on the mushroom farmer for the entire class. After students watch the video (3:46 minutes), prompt them to reflect on the profession by answering the two questions independently.

TEACHER MOVES

Consider beginning to incorporate language of growth mindset, taking initiative or risk, and other key characteristics of your classroom culture you want to build.

Identify which qualities are important for you and your students and use this video as a starting point. "How does Alejandro show _________?"

1.1.1 Warm-Up: Day in the Life of a Mushroom Farmer

1. Alejandro, a mushroom-kit entrepreneur, says that when it comes to growing his business, "The most important thing is … adapting opportunities that come to you and still holding the same amount of passion."

   a. What is something you are passionate about?
   Answers will vary. I am passionate about cooking and playing sports.

   b. How has a new interest or hobby surprised you?
   Answers will vary. A new interest that surprised me was creating Tik Tok videos because I am normally shy about dancing in front of people.

1.1.2 REFRESHER: DEFINING A STATISTICAL QUESTION

TEACHER MOVES

Students may state that "What time does school end?" is a statistical question. Query them to explain their thinking. If the question is posed to students at different schools, the question has variability. If the question is posed to students at the same school, it does not.

ENRICHMENT

Students may pause when describing what is meant by "variability in the data." Consider the following questions for scaffolding:

- What does 'variability' mean?
- What is the root word of 'variable'?
- How many responses can statistical questions have?
- Which questions would have responses that vary?

1.1.2 Refresher: Defining a Statistical Question

A ***statistical question*** is a question that can be answered by collecting data. Often, there will be variability in the data.

Examples and non-examples of statistical questions are shown.

Statistical Questions	NOT Statistical Questions
• How many siblings do the students in my class have? • How tall are the players on the girls' basketball team? • What is the favorite song of the 6th graders at my school?	• What time does school start? • Who is the President of the United States? • What is Avery's favorite color?

1. What do you notice about the two types of questions?
   Answers will vary. Statistical questions provide variability. The answers to statistical questions will vary depending on who is answering the questions. The answers to non-statistical questions have one response.

2. What do you think "variability in the data" means in the definition of "statistical question?"
   Answers will vary. Variability in the data means that there are various data values anticipated.

3. Explain why the question, "How many times do the students at my school go swimming this summer?" is a statistical question.
   Answers will vary. This question is a statistical question because it requires you to collect data. There will be variability in the answers to this question.

1.1.3 Try It: Identifying Statistical Questions

1. Sort the following questions into the appropriate column by writing the letters of the questions in the table provided.

   a. How many miles do the students in my class live from the school campus?
   b. How many counties are there in Florida?
   c. Who is the governor of Florida?
   d. How many siblings do my 7th grade teachers have?
   e. How many hours do adults in Florida spend outdoors on a weekend?
   f. How many days are in December?

Statistical Questions	NOT Statistical Questions
A, D, E	*B, C, F*

1.1.3 TRY IT: IDENTIFYING STATISTICAL QUESTIONS

TEACHER MOVES

This could be a whole-group activity by having students volunteer to read each question and afterward polling the class on whether it is a statistical question or not. Encourage students to share their thoughts about the choices they made.

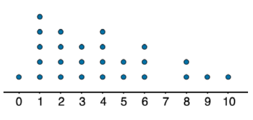
1.1.4 Refresher: Data Displays

Numerical data can be displayed graphically in a variety of ways. Three graphs that are commonly used are histograms, line plots, and box plots.

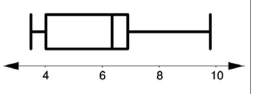
1. Three data displays are shown below. Match each type of data display with the examples shown by writing the letter on the line.

   Histogram *C*   Line Plot *A*   Box Plot *B*

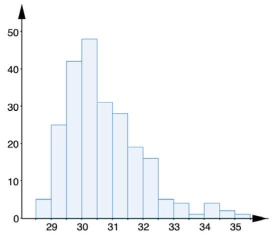
Data Display A



Data Display B



Data Display C



1.1.4 REFRESHER: DATA DISPLAYS

ENRICHMENT

Prompt students to begin thinking about how to use and interpret each data display in preparation for 1.1.5 Guided Instruction by asking questions like:

- Could all three displays be used to show the same data? Why or why not?
- Could any two be used to show the same data? Which two? Why?

1.1.5 GUIDED INSTRUCTION: INTERPRETING HISTOGRAMS AND LINE PLOTS

TEACHER MOVES

Begin by introducing the histogram that will be used. For the first question, allow students to work through it individually before engaging in a discussion around the questions. If students struggle with the concept of 'skewed' and 'outlier,' encourage them to give their best answer before the discussion. Consider using some of the prompts outlined below. Students could work with a partner to complete question 2.

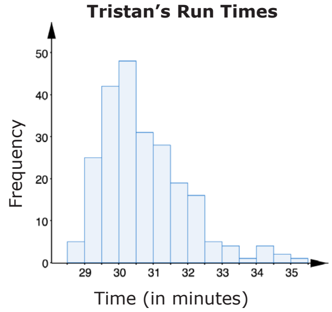
- •What does it mean if something has symmetry? Is the histogram symmetrical? Why or why not?
- •What does it mean if data is 'skewed'?
- •What types of things can cause data to be skewed?
- •Where does most of the data appear to fall in the histogram?
- •Does the tail of the data trail off to the left or to the right of the peak?

QUESTIONING

For a challenge, have students consider and explain how the shape of the histogram might change if Tristan recorded more run times that all fell below the 30- to 31-minute mark.

1.1.5 Guided Instruction: Interpreting Histograms and Line Plots

1. Tristan has been tracking the time, in minutes, it takes him to run 4 miles for the last 10 years. A histogram of Tristan's run times is shown.



a. The shape of run times can be best described as

○ symmetrical.
○ right skewed.
○ left skewed.

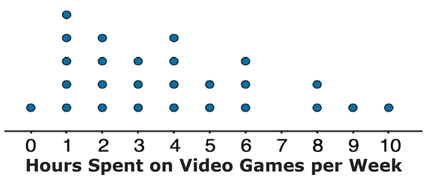
b. Tristan's typical run time was about 30 minutes. Most of his run times were between 29.5 to 30.5 minutes. Not many of his runs were

○ shorter
○ longer

than 32.5 minutes.

c. If the data value ○ 28 ○ 33 ○ *48* was added to the histogram, it would be considered an outlier.

2. The line plot shows the number of hours per week that students reported playing video games.



a. How many students were surveyed?
26 students

b. How many students reported spending 1 hour on playing video games each week?
5 students

c. What percentage of the students reported spending 4 hours or fewer on video games each week?
65% or 65.38%

d. What is the range of time spent playing video games in this group?
10 hours

1.1.5 GUIDED INSTRUCTION: INTERPRETING HISTOGRAMS AND LINE PLOTS (CONTINUED)

DIFFERENTIATE INSTRUCTION

Consolidating all the different vocabulary terms addressed in the lesson may support students as they analyze and interpret the data presented. Consider using a foldable or some other type of visual to provide support for both auditory and visual learners.

TEACHER MOVES

Students may have forgotten how to calculate a percentage.As this is not an outcome of this lesson, prompting them with the question, ‘What is 5% of 26?’ and asking them to write it as an equation might provide the scaffolding to successfully complete question 2c.

1.1.6 COLLABORATION: WHICH GRAPH?

TEACHER MOVES

Consider introducing the task by engaging students in a Compare and Contrast exercise as they answer 1a to prompt them to think about the different characteristics of displays and how these might be used to help interpret and analyze data. Partners then work together to answer 1b through 1f.

Compare and Connect

Consider engaging students in Compare and Connect to begin the collaboration to prompt students to think about how different displays are used to show data.

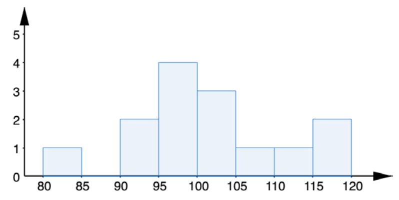
Engage them in thinking and talking about the graphs by demonstrating a Think Aloud:

- What does a histogram tell me about a given data set? (It does tell me how many, but it's given in intervals rather than individual data points.)
- What does the bar representing the interval between 95 and 100 tell me about the lengths of the Pixar films sampled? (There are 4 that are between 95 and 100 minutes long.)
- Can the histogram tell me which length was the most common? (Because the data is presented in intervals, I can't really tell which length was the most common.)

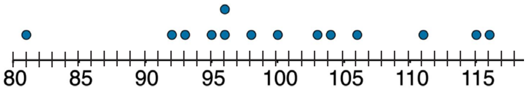
1.1.6 Collaboration: Which Graph?

1. A movie critic recorded the length, in minutes, of a sample of Pixar films and made a histogram and a line plot of the data.

Film Length



Length (in minutes)



Length (in minutes)

a. Which graph could tell you how many film lengths were recorded?
Both graphs can be used to determine how many film lengths were recorded but the Line Plot is easier to make this determination.

b. How many film lengths are recorded?
14

c. How many films lasted at least 95 minutes but fewer than 100 minutes, according to the histogram?
4

d. How many films lasted at least 95 minutes but fewer than 100 minutes, according to the line plot?
4

e. What is the range of the film lengths?
35 minutes

f. Work with your partner to explain which graph was used to determine the range.
The line plot was used because it clearly shows the length of the longest and shortest films.

g. Are there any outliers? If there are, name them.
81 minutes is the outlier in this data set.

1.1.6 COLLABORATION: WHICH GRAPH? (CONTINUED)

TEACHER MOVES

Students may identify the range to be "Between 81 and 115 minutes long." Remind them that a range is the difference between the highest and lowest values in a data set and is represented as a whole number.

1.1.7 Wrap-Up: Reflection

1. Choose the emoji that reflects your understanding of statistics and interpreting data.

Answers will vary.

1.1.7 WRAP-UP: REFLECTION

TEACHER MOVES

Consider asking students to identify one skill they struggled with or one skill they felt they understood very well and could explain to others.